

M003 Extension 2 — English Worksheet (Beginner)

Topic: 3D Mob Heads (x, y, z) · **Course:** 3D Coordinates · **Level:** Beginner (Extension, after Week 3) ·

Time: about 30 minutes

Build a mob head in 3D. Every block needs **three** numbers: **(x, y, z)**. x is **across**, y is **up**, z is **deeper** (forward). The head stands in front of you. Look at it as you build.

● Stand on your **home spot** (red block) every run. Move, and the head moves too.

1 · Predict 🧠

Read the code. Do **not** run it. Circle one word.

```
place green block at (3, 5, 1)
place green block at (3, 5, 2)
place green block at (3, 5, 3)
```

Only z changes. The line goes: across · up · deeper


```
place green block at (4, 5, 1)
place green block at (4, 6, 1)
place green block at (4, 7, 1)
```

Only y changes. The line goes: across · up · deeper

2 · Spot the Bug 🐛

This eye should be one step **deeper**, at (4, 7, 2). The code below only has **two** numbers, so it lands on the front.

```
# broken  
place black block at (4, 7)
```

Fill the blank to fix it:

```
place black block at (4, 7, __)
```

Word bank: 2

Write one sentence: what was missing?

3 · Build It

Look at the creeper head in the world. Walk around it.

Build a small solid head — a cube. Keep it easy:

Front face: fill green blocks from $(3, 5, 1)$ up to $(5, 7, 1)$ — a small square.

Add depth: build the same square again at $z = 2$, then $z = 3$.

Add 2 eyes on the front only: black block at (3, 7, 1) and black block at (5, 7, 1).

💡 Tip: the eyes only show on the **front**. The sides and back are plain green.

Send a photo. Then finish these 2 sentences.

I built the ... head. It is ... blocks deep.

To go deeper I changed the ... number.

4 · Explain It

Film a short video. Show these 3 things:

1. Show your home spot.
2. Point at the front face and one eye.
3. Show how deep your head is ($z = 1, 2, 3$).

Write 2 short sentences about your head:

Words to use: **x · y · z · deeper**

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.